

Bhalaji Yadav Kantepalle

Richmond, VA | | contact@bhalaji.com
linkedin.com/in/kbhalajiyadav | Portfolio: bhalaji.com

PROFESSIONAL SUMMARY

Chemical Engineer (M.S.) and Researcher specializing in **Soft Materials Mechanics**, **Optical Metrology**, and **Computational Physics**. Expert in developing "ground truth" validation systems using **Python (OpenCV)** and **Fracture Mechanics** principles. Proven ability to characterize complex material behavior (Adhesion, Fatigue, Rheology) and translate findings into scalable engineering solutions. Successfully secured \$30,000 in research funding through technical proposal writing.

EDUCATION

Virginia Commonwealth University (VCU)

M.S., Chemical & Life Science Engineering | GPA: 3.62

Richmond, VA

Exp. Dec 2026

Core Research: Fracture Mechanics of Soft Interfaces, Mechanochromic Sensors, Computer Vision for Metrology.

Achievement: Graduate Assistantship Award (Fall 2025) – Full Tuition Waiver & Stipend.

Amrita University

B.Tech., Chemical Engineering

Coimbatore, India

Jun 2022

TECHNICAL SKILLS

Materials Characterization: Fracture Mechanics, Adhesion Testing (ASTM D2724), Rheology, Optical Microscopy, Tensile Testing

Computational R&D: Python (OpenCV, Pandas, Scipy), Image Processing, CIE L*a*b* Colorimetry, Data Modeling

Lab Instrumentation: Universal Testing Machine (TA.XTplusC Texture Analyser), NMR Spectroscopy, FT-NMR, UV-Vis, 3D Printing

Engineering Standards: Experimental Design (DOE), Root Cause Analysis (RCA), Technical Grant Writing, FDA cGMP

RESEARCH EXPERIENCE

VCU Soft Functional Materials Lab

Graduate Researcher & Metrology Lead

Richmond, VA

Sep 2024 – Present

- Novel Optical Metrology:** Engineered a high-throughput **Computer Vision pipeline** (Python/OpenCV) to quantify mechanochromic response in smart textiles. Synchronized sub-pixel texture tracking with CIE L*a*b* colorimetry to eliminate lighting artifacts.
- Fracture Mechanics Automation:** Developed a first-principles analysis protocol to identify "false positive" adhesion failures in soft wearables. Automated the extraction of **Peel Stability Index (PSI)** metrics, reducing data processing time by >80%.
- Grant Success (\$30,000):** Drafted a successful research proposal for the Commonwealth Cyber Initiative. Synthesized preliminary data to substantiate the clinical viability of thermochromic sensors for diabetic health monitoring.
- Material Validation:** Characterized the viscoelastic properties of complex polymer fluids using **Rheology models** (Cross/Power-law) to optimize coating formulations for scalable manufacturing.

INDUSTRY EXPERIENCE

Kreative Organics Pvt. Ltd.

Technical Project Manager

Hyderabad, India

May 2023 – May 2024

- Data Engineering:** Designed and containerized a **Python/Docker** data protocol to standardize chemical nomenclature across global supply chains, improving data retrieval accuracy by 90%.
- Validation Logic (FDA):** Managed the **SAP ERP Re-qualification**, partnering with consultants to design rigorous OQ/PQ testing protocols that ensured compliance with FDA data integrity standards.
- Process Optimization:** Applied engineering principles to optimize system cutover logistics, reducing downtime from 5 days to 5 hours through precise critical-path planning.

Technical Project Intern

Oct 2022 – Apr 2023

- R&D Workflow Automation:** Engineered a digital tool that analyzed global trade datasets, reducing manual research time by 70% and streamlining the identification of viable chemical targets and markets.

PUBLICATIONS & ACHIEVEMENTS

Journal Article: I. Caloian, J. Trapp, B. Y. Kantepalle, et al. "Mechanical Properties of **Dual-Layer Electrospun Fiber Mats**". *Polymers* 17(13), 1777 (2025). [doi:10.3390/polym17131777]

Selected Participant: 1st National Neutron Scattering School, Oak Ridge National Laboratory (Sep 2025).

Poster Presentation: "Adhesives for Personalized Wearable Devices" (ACS Fall 2025, Washington D.C.).